

Computing and Software
SFWRENG 2C03
Data Structures and Algorithms
Winter 2025



ENGINEERING

Instructor Information

Samuel Scott

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Office: ETB 111

Office Hours:

Wednesday, 11:30 to 1:30

Class Times

Lectures

- Tuesday, Wednesday, Friday 9:30 - 10:20, BSB B135

Tutorials

- T01: Thursday, 12:30 - 1:20, BSB 104
- T02: Tuesday, 11:30 - 12:20, BSB 105
- T03: Thursday, 1:30 - 2:20, ITB 139
- T04: Thursday, 1:39 - 2:20, ABB 166

Class Format

This is an In Person class (3 hours lecture, 1 hour tutorial). Attendance and in-person participation in lecture and tutorial activities is strongly recommended.

Course Dates: 01/06/2025 - 04/08/2025

Units: 3.00

Course Delivery Mode: In Person

Course Description: Basic data structures: stacks, queues, hash tables, and binary trees; searching and sorting; graph representations and algorithms, including minimum spanning trees, traversals, shortest paths; introduction to algorithmic design strategies; correctness and performance analysis. Implementation of data structures and algorithms. Three lectures, one tutorial (one hour); second term Prerequisite(s): SFWRENG 2DM3; registration in any Software Engineering program Antirequisite(s): COMPSCI 2C03, COMPENG 3SM4

Important Links

- [Mosaic](#)
- [Avenue to Learn](#)
- [Student Accessibility Services - Accommodations](#)
- [McMaster University Library](#)
- [eReserves](#)

- [Computing and Software Drop-in Help](#)

- [Equity and Inclusion](#)
- [2SLGBTQIA+ Resources](#)
- [Black Lives Matter](#)
- [Indigenous Student Services](#)
- [MES Clubs](#)

- [Counselling and Well Being](#)
- [Student Success Centre](#)

- [Academic Advising](#)
- [Student Technical Support](#)
- [Google Chrome Accessibility](#)

Course Learning Outcomes

- 1. Describe the properties, common implementations, and use-cases of well-known data structures (stacks, queues, trees, hash tables, graphs, etc.)
- 2. Implement established sorting, searching, tree, and graph algorithms using pseudo-code and/or code in a high level programming language..
- 3. Design “client” algorithms that could be used to solve problems or implement use-cases with stacks, queues, trees, hash tables, and graphs.
- 4. Evaluate the time complexity (best-, average-, and worst-case running times) of various algorithms through algorithm analysis and experimentation.
- 5. Identify the time and space trade-offs in implementing data structures and algorithms.
- 6. Prove basic properties of various data structures, algorithms, and complexity classes.
- 7. Justify the selection of data structures and algorithms to solve a variety of computing problems.

Graduate Attributes

The Canadian Engineering Accreditation Board (CEAB) is a division of Engineers Canada and is responsible for accrediting undergraduate engineering programs across Canada. Accreditation by the CEAB ensures that the engineering programs meet a national standard of quality and cover essential educational requirements. Graduate Attributes are a set of qualities and skills that the CEAB expects engineering graduates

to possess. These attributes are a benchmark for the learning outcomes of accredited engineering programs. This section lists the Graduate Attribute Indicators associated with the Learning Outcomes in this course.

- Graduate Attribute Indicator 1.3: Competence in Engineering Fundamentals (Learning Outcome 1)
- Graduate Attribute Indicator 1.4: Competence in Specialized Engineering knowledge (Learning Outcome 2)
- Graduate Attribute Indicator 2.1: Identifies and states reasonable assumptions and suitable engineering fundamentals, before proposing a solution path to a problem (Learning Outcome 3)
- Graduate Attribute Indicator 2.2: Proposes problem solutions supported by substantiated reasoning, recognizing the limitations of the solutions (Learning Outcome 5)
- Graduate Attribute Indicator 3.1: Selects appropriately from relevant knowledge base to plan appropriate data collection methods and analysis strategies (Learning Outcome 6)
- Graduate Attribute Indicator 4.1: Defines the problem by identifying relevant context, constraints, and prior approaches before exploring potential design solutions (Learning Outcome 3)
- Graduate Attribute Indicator 4.2: Explores a breadth of potential solutions, considering their benefits and trade-offs as they relate to the project requirements (Learning Outcome 5)
- Graduate Attribute Indicator 7.1: Demonstrates comprehension of technical and non-technical instructions and questions (Learning Outcome 2)
- Graduate Attribute Indicator 7.2: Composes an effective written document for the intended audience (Learning Outcome 7)

Course Schedule

Week	Dates	Content	Textbook	Evaluation
Linear Data Structures and Sorting				
1	Jan 7, 8, 10	Introductions, Abstract Data Types, 1.2, 1.3		

		Linked Lists		
2	14, 15, 17	Stacks, Queues, Basic Sorting	1.3, 2.1	Assignment 1
3	21, 22, 24	Algorithm Analysis and Merge Sort	1.4, 2.2	
4	28, 29, 31	Quick Sort, Radix Sort	2.3, 5.1	Assignment 2
Trees and Searching				
5	Feb 4, 5, 7	Midterm, Symbol Tables, Binary Search Trees	3.1	Midterm 1, Feb 4 PGCLL B138
6	11, 12, 14	Balanced Trees	3.2	
BREAK WEEK				
7	25, 26, 28	Balanced Trees, Heaps, Heap Sort	3.3, 2.4	Assignment 3
8	Mar 4, 5, 7	Hash Tables	2.4, 3.4	
9	11, 12, 14	Midterm, Graphs		Midterm 2, Mar 11 PGCLL B138
Graphs and Path Finding				
10	18, 19, 21	Graphs, Digraphs	4.1, 4.2	
11	25, 26, 28	Digraphs, Edge-Weighted Diagraphs	4.2, 4.3	Assignment 4
12	Apr 1, 2, 4	Minimum Spanning Trees, Shortest Paths	4.3, 4.4 (1.5)	
13	8	Review		
EXAM WEEK (Final Exam)				

Required Materials and Texts

Textbook Listing: <https://textbooks.mcmaster.ca>

Algorithms, 4th Edition

Authors: R. Sedgewick and K Wayne

Publisher: Addison-Wesley

Publication Date: 2011

Course Evaluation

Assignments

- 30% (4, equally weighted)

Midterm Exams

- 30% (2, equally weighted)

Final Exam

- 40%

Course Evaluation Details

Assignments

Each assignment will consist of paper-and-pencil coding exercises related to the most recently covered content. This could involve writing code or pseudocode, designing algorithms, analyzing algorithms, mathematical proofs, derivations, or other written exercises. The four assignments are equally weighted.

You are encouraged to discuss the course material with your classmates, but your assignments must be your own individual effort. You may consult other sources, but all such sources must be documented. Failure to do so may result in academic dishonesty charges.

Exams

All exams will be closed-book, paper-and-pencil tests. There will be two midterms written during class in weeks 5 and 9 (check the schedule above for the date your tutorial section writes the test). The midterms will cover the most recent material from the lectures and tutorials. The final exam will be scheduled after the end of classes. The final exam will be cumulative.

Grading Scale

The McMaster 12 Point Grading Scale

Grade	Equivalent Grade Point	Equivalent Percentages
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Grade	Equivalent Grade Point	Equivalent Percentages
A+	12	90-100
A	11	85-89
A-	10	80-84
B+	9	77-79
B	8	73-76
B-	7	70-72
C+	6	67-69
C	5	63-66
C-	4	60-62
D+	3	57-59
D	2	53-56
D-	1	50-52
F	0	0-49

Late Assignments

Assignments are due on the day listed on the assignment handout (i.e., up to 11:59:59 pm that day). After that, they are late. Late days are tracked by calendar date. For example, if the due date is November 1st and you hand the assignment in some time on November 2nd, it is one day late.

You can hand in late assignments with no penalty, but you have a limit of 8 late days for all assignments combined. If you go over that limit, late assignments will no longer be accepted. Please note that the final day to hand in any work for this course, regardless of how many late days you have remaining, will be the Thursday of Week 13.

Your 8 late days do not include special or unforeseen circumstances (illnesses, family emergencies, etc.). If you are delayed in completing an assignment due to special or unforeseen circumstances, let your professor know as soon as you can so that you can negotiate a new due date. Keep in mind that you may be required to produce supporting documentation.

Absences, Missed Work, Illness

Assignments

Your 8 late days do not include special or unforeseen circumstances (illnesses, family emergencies, etc.). If you are delayed in completing an assignment due to special or unforeseen circumstances, let your professor know as soon as you can so that you can negotiate a new due date. Keep in mind that you may be required to produce supporting documentation. If you MSAF an assignment, we will either negotiate a new due date or move the weight of the assignment to the final exam.

Midterm Exams

If you miss the midterm exam for a legitimate reason (MSAF), the weight of that exam will be moved to the final exam.

Generative AI: Use Prohibited

Students are not permitted to use generative AI in this course. In alignment with [McMaster academic integrity policy](#), it “shall be an offence knowingly to ... submit academic work for assessment that was purchased or acquired from another source”. This includes work created by generative AI tools. Also state in the policy is the following, “Contract Cheating is the act of “outsourcing of student work to third parties” (Lancaster & Clarke, 2016, p. 639) with or without payment.” Using Generative AI tools is a form of contract cheating. Charges of academic dishonesty will be brought forward to the Office of Academic Integrity.

APPROVED ADVISORY STATEMENTS

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on

the transcript (notation reads: “Grade of F assigned for academic dishonesty”), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/), located at <https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/>

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one’s own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Courses with an On-line Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online.**

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

Equity, Diversity, and Inclusion

The Faculty of Engineering is committed to creating an environment in which students of all genders, cultures, ethnicities, races, sexual orientations, abilities, and socioeconomic backgrounds have equal access to education and are welcomed and treated fairly. If you have any concerns regarding inclusion in our Faculty, in particular if you or one of your peers is experiencing harassment or discrimination, you are encouraged to contact the Chair, Associate Undergraduate Chair, Academic Advisor or to contact the [Equity and Inclusion Office](#).

Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University’s [Academic Accommodation of Students with Disabilities](#) policy.

Academic Advising

For any academic inquiries please reach out to the Office of the Associate Dean (Academic) in Engineering located in JHE-Hatch 301.

Details on academic supports and contact information are available from:

<https://www.eng.mcmaster.ca/programs/academic-advising>

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

Academic Accommodation for Religious, Indigenous, or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office ***normally within 10 working days*** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.

Turnitin.com

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.